

CSD19536KCS 100V N-Channel NexFET™ Power MOSFET

1 Features

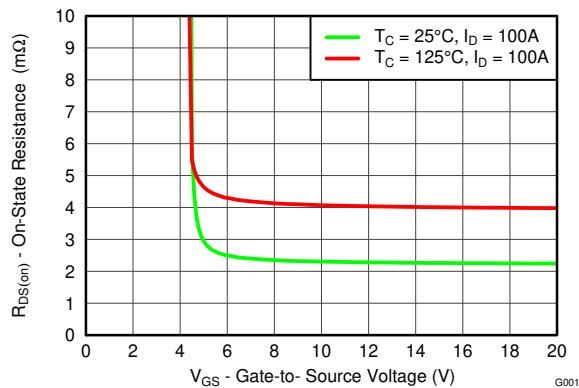
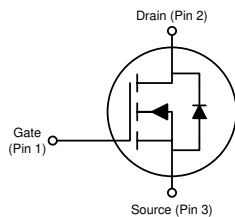
- Ultra-low Q_g and Q_{gd}
- Low thermal resistance
- Avalanche rated
- Pb-Free terminal plating
- RoHS compliant
- Halogen free
- TO-220 plastic package

2 Applications

- Secondary side synchronous rectifier
- Motor control

3 Description

This 100V, 2.3m Ω , TO-220 NexFET™ power MOSFET is designed to minimize losses in power conversion applications.



$R_{DS(on)}$ vs V_{GS}

Product Summary

$T_A = 25^\circ\text{C}$		TYPICAL VALUE		UNIT
V_{DS}	Drain-to-Source Voltage	100		V
Q_g	Gate Charge Total (10V)	118		nC
Q_{gd}	Gate Charge Gate to Drain	17		nC
$R_{DS(on)}$	Drain-to-Source On-Resistance	$V_{GS} = 6\text{V}$	2.5	m Ω
		$V_{GS} = 10\text{V}$	2.3	m Ω
$V_{GS(th)}$	Threshold Voltage	2.5		V

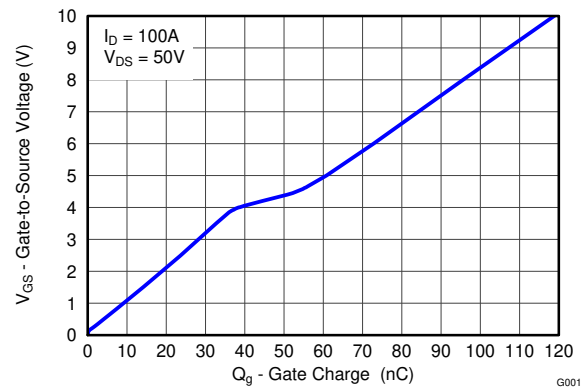
Ordering Information

Device	Package	Media	Qty	Ship
CSD19536KCS	TO-220 Plastic Package	Tube	50	Tube

Absolute Maximum Ratings

$T_A = 25^\circ\text{C}$		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage	100	V
V_{GS}	Gate-to-Source Voltage	± 20	V
I_D	Continuous Drain Current (Package limited)	150	A
	Continuous Drain Current (Silicon limited), $T_C = 25^\circ\text{C}$	259	
	Continuous Drain Current (Silicon limited), $T_C = 100^\circ\text{C}$	183	
I_{DM}	Pulsed Drain Current ⁽¹⁾	400	A
P_D	Power Dissipation	375	W
T_J, T_{stg}	Operating Junction and Storage Temperature Range	-55 to 175	$^\circ\text{C}$
E_{AS}	Avalanche Energy, single pulse $I_D = 127\text{A}, L = 0.1\text{mH}, R_G = 25\Omega$	806	mJ

(1) Max $R_{\theta JC} = 0.4^\circ\text{C/W}$, pulse duration $\leq 100\mu\text{s}$, duty cycle $\leq 1\%$



Gate Charge



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4 Specifications

4.1 Electrical Characteristics

(T_A = 25°C unless otherwise stated)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	
STATIC CHARACTERISTICS							
BV _{DSS}	Drain-to-Source Voltage	V _{GS} = 0V, I _D = 250μA	100			V	
I _{DSS}	Drain-to-Source Leakage Current	V _{GS} = 0V, V _{DS} = 80V			1	μA	
I _{GSS}	Gate-to-Source Leakage Current	V _{DS} = 0V, V _{GS} = 20V			100	nA	
V _{GS(th)}	Gate-to-Source Threshold Voltage	V _{DS} = V _{GS} , I _D = 250μA	2.1	2.5	3.2	V	
R _{DS(on)}	Drain-to-Source On-Resistance	V _{GS} = 6V, I _D = 100A			2.5	3.2	mΩ
		V _{GS} = 10V, I _D = 100A			2.3	2.7	mΩ
g _{fs}	Transconductance	V _{DS} = 10V, I _D = 100A	307			S	
DYNAMIC CHARACTERISTICS							
C _{iss}	Input Capacitance	V _{GS} = 0V, V _{DS} = 50V, f = 1MHz		9250	12000	pF	
C _{oss}	Output Capacitance			1820	2370	pF	
C _{rss}	Reverse Transfer Capacitance			47	61	pF	
R _G	Series Gate Resistance	V _{DS} = 50V, I _D = 100A		1.4	2.8	Ω	
Q _g	Gate Charge Total (10V)			118	153	nC	
Q _{gd}	Gate Charge Gate to Drain			17		nC	
Q _{gs}	Gate Charge Gate to Source			37		nC	
Q _{g(th)}	Gate Charge at V _{th}			24		nC	
Q _{oss}	Output Charge	V _{DS} = 50V, V _{GS} = 0V		335		nC	
t _{d(on)}	Turn On Delay Time	V _{DS} = 50V, V _{GS} = 10V, I _{DS} = 100A, R _G = 0Ω		14		ns	
t _r	Rise Time			8		ns	
t _{d(off)}	Turn Off Delay Time			38		ns	
t _f	Fall Time			5		ns	
DIODE CHARACTERISTICS							
V _{SD}	Diode Forward Voltage	I _{SD} = 100A, V _{GS} = 0V		0.9	1.1	V	
Q _{rr}	Reverse Recovery Charge	V _{DS} = 50V, I _F = 100A, di/dt = 300A/μs		548		nC	
t _{rr}	Reverse Recovery Time			110		ns	

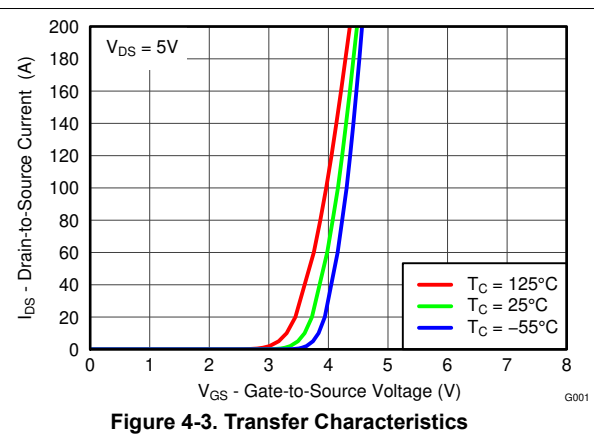
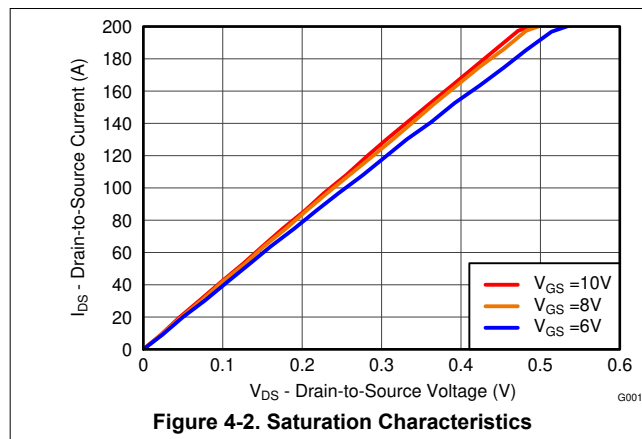
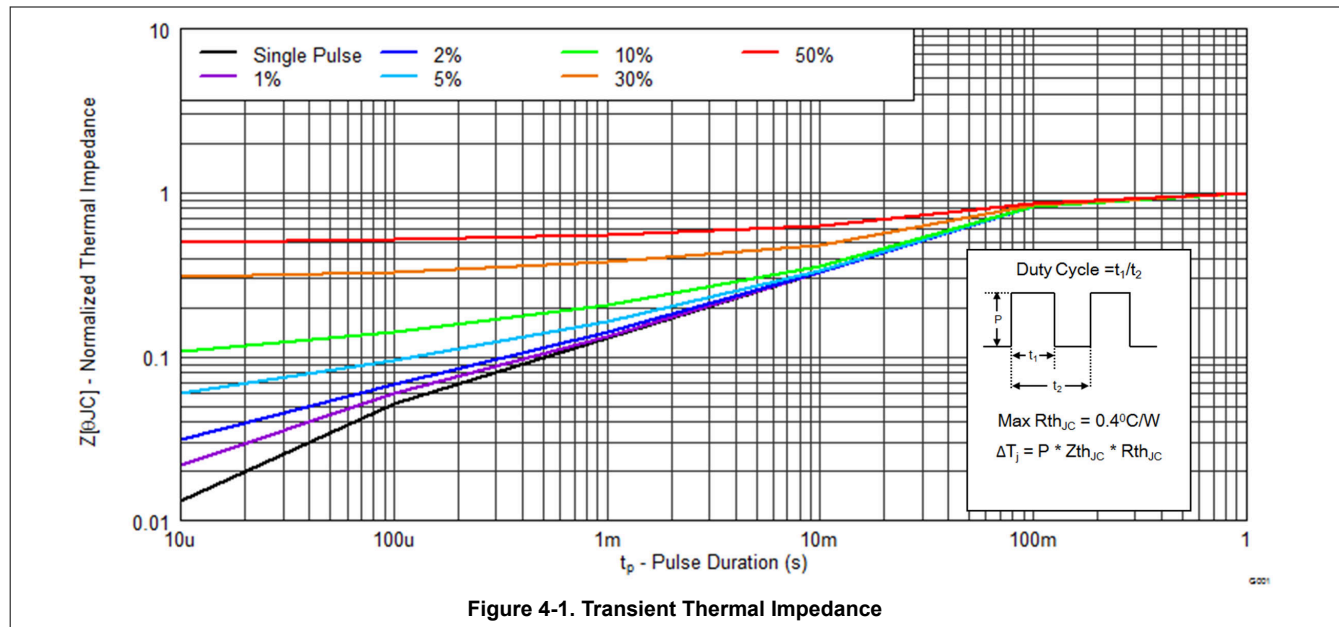
4.2 Thermal Information

(T_A = 25°C unless otherwise stated)

THERMAL METRIC		MIN	TYP	MAX	UNIT
R _{θJC}	Junction-to-Case Thermal Resistance			0.4	°C/W
R _{θJA}	Junction-to-Ambient Thermal Resistance			62	

4.3 Typical MOSFET Characteristics

($T_A = 25^\circ\text{C}$ unless otherwise stated)



4.3 Typical MOSFET Characteristics (continued)

($T_A = 25^\circ\text{C}$ unless otherwise stated)

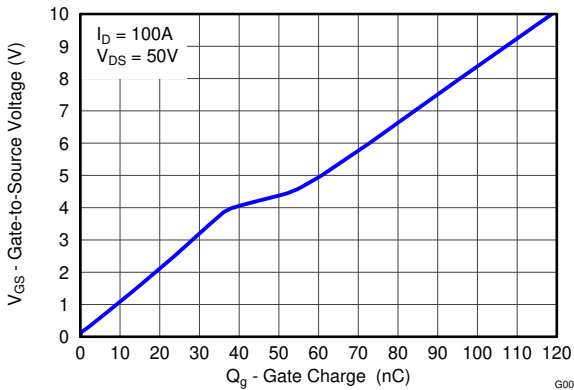


Figure 4-4. Gate Charge

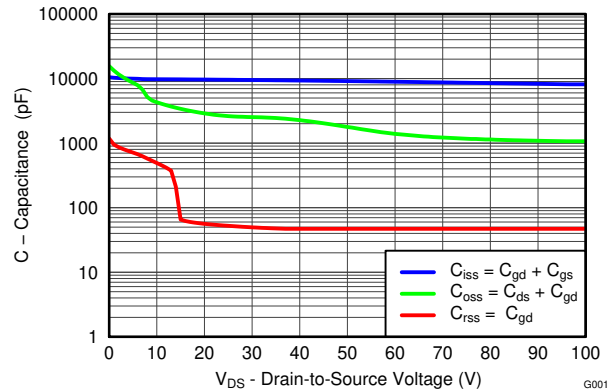


Figure 4-5. Capacitance

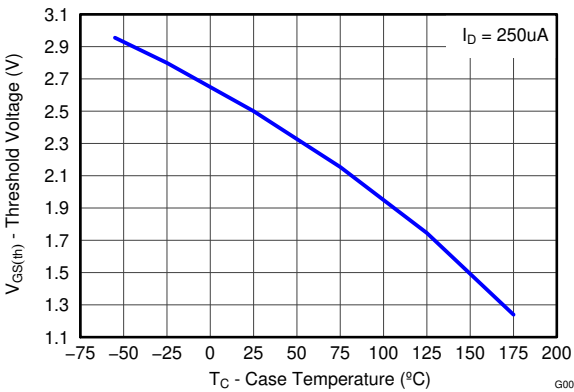


Figure 4-6. Threshold Voltage vs Temperature

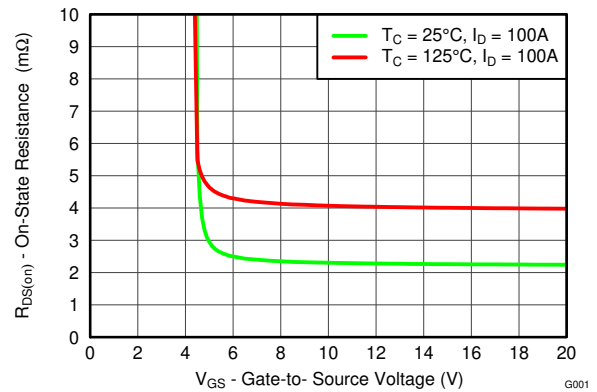


Figure 4-7. On-State Resistance vs Gate-to-Source Voltage

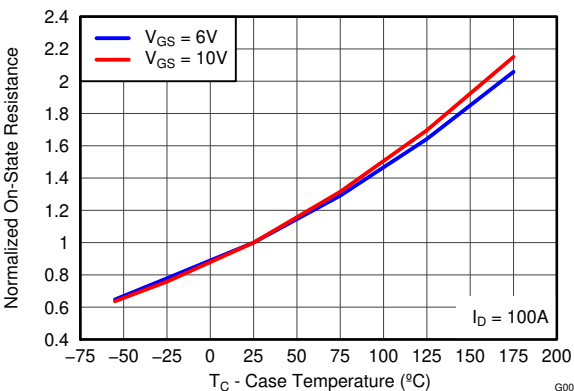


Figure 4-8. Normalized On-State Resistance vs Temperature

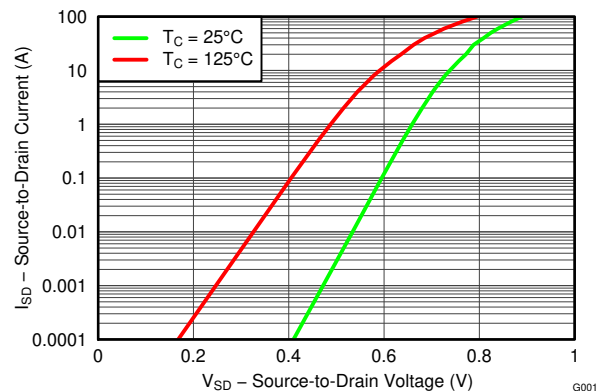


Figure 4-9. Typical Diode Forward Voltage

4.3 Typical MOSFET Characteristics (continued)

($T_A = 25^\circ\text{C}$ unless otherwise stated)

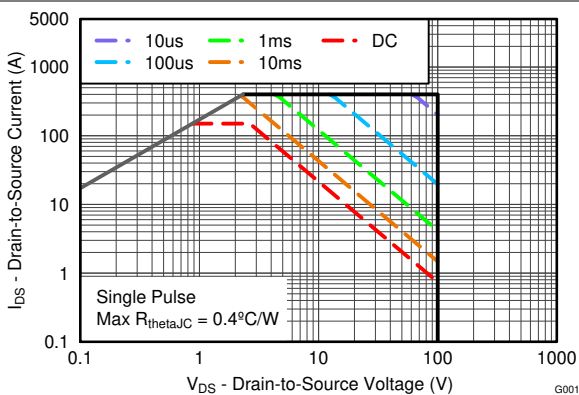


Figure 4-10. Maximum Safe Operating Area

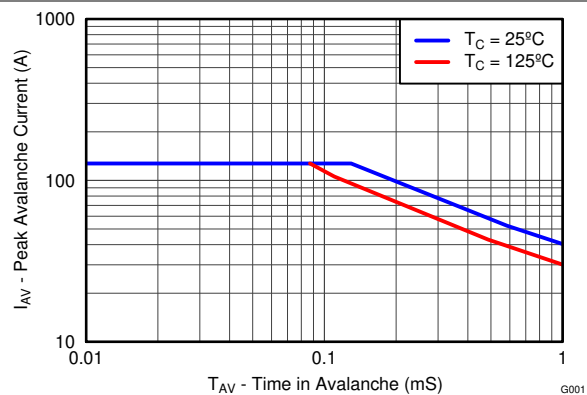


Figure 4-11. Single Pulse Unclamped Inductive Switching

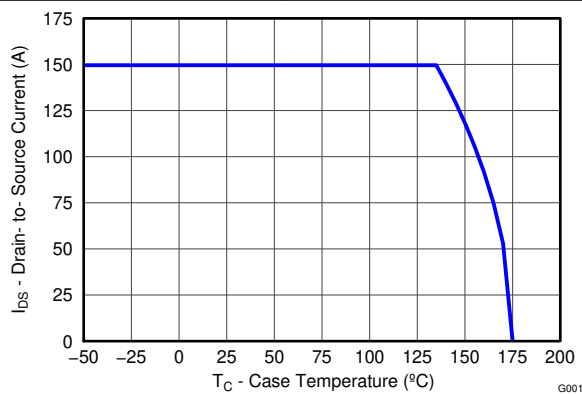


Figure 4-12. Maximum Drain Current vs Temperature

5 Device and Documentation Support

5.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on [ti.com](https://www.ti.com). Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

5.2 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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5.3 Trademarks

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5.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

5.5 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

6 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision B (October 2014) to Revision C (May 2024) Page

- Updated the numbering format for tables, figures, and cross-references throughout the document..... 1

Changes from Revision A (April 2014) to Revision B (April 2014) Page

- Updated Pulsed Drain Current conditions 1
- Updated the SOA in [Figure 4-10](#) 4

Changes from Revision * (January 2014) to Revision A (April 2014) Page

- Increased pulsed current rating to 400A 1
- Updated SOA curve..... 4

7 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CSD19536KCS	Active	Production	TO-220 (KCS) 3	50 TUBE	ROHS Exempt	SN	N/A for Pkg Type	-55 to 175	CSD19536KCS

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

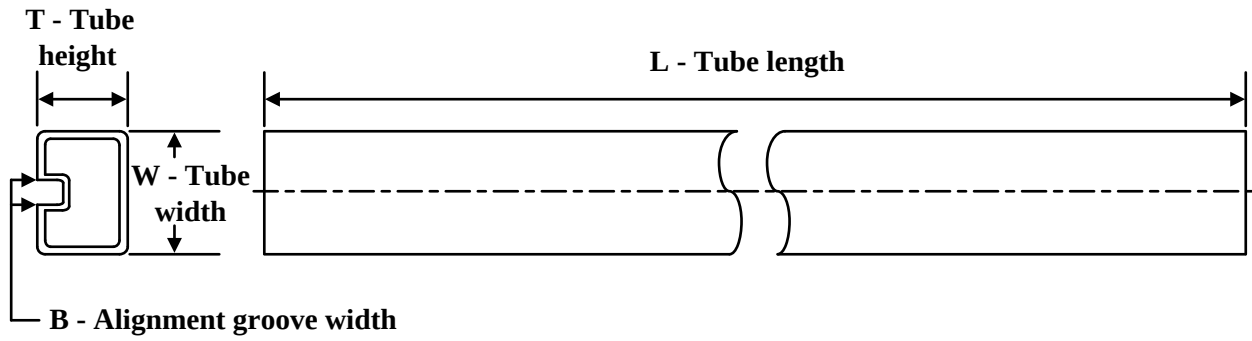
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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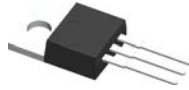
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
CSD19536KCS	KCS	TO-220	3	50	532	34.1	700	9.6
CSD19536KCS	KCS	TO-220	3	50	534.5	33	7000	3.4

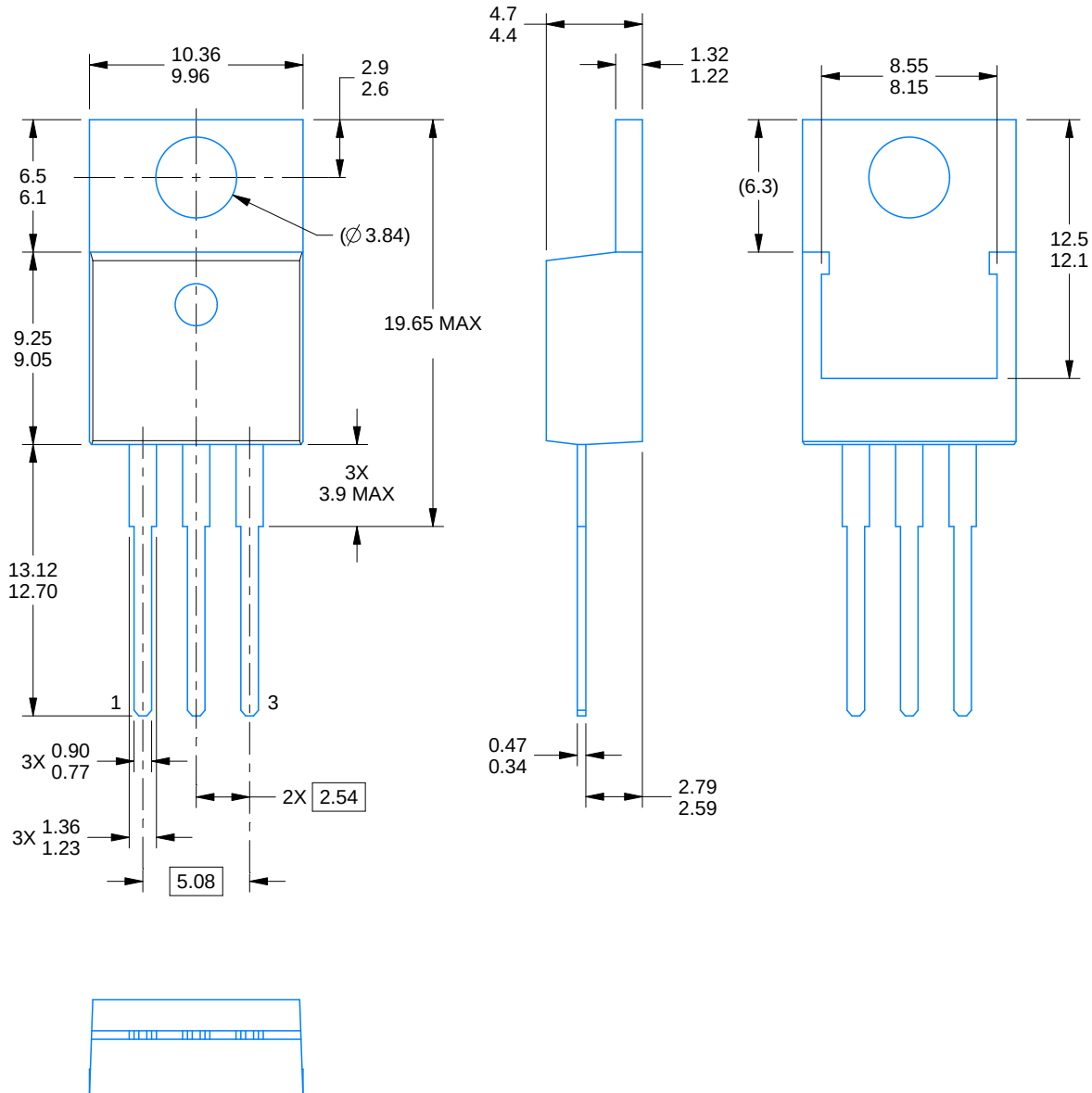


PACKAGE OUTLINE

KCS0003B

TO-220 - 19.65 mm max height

TO-220



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NOTES:

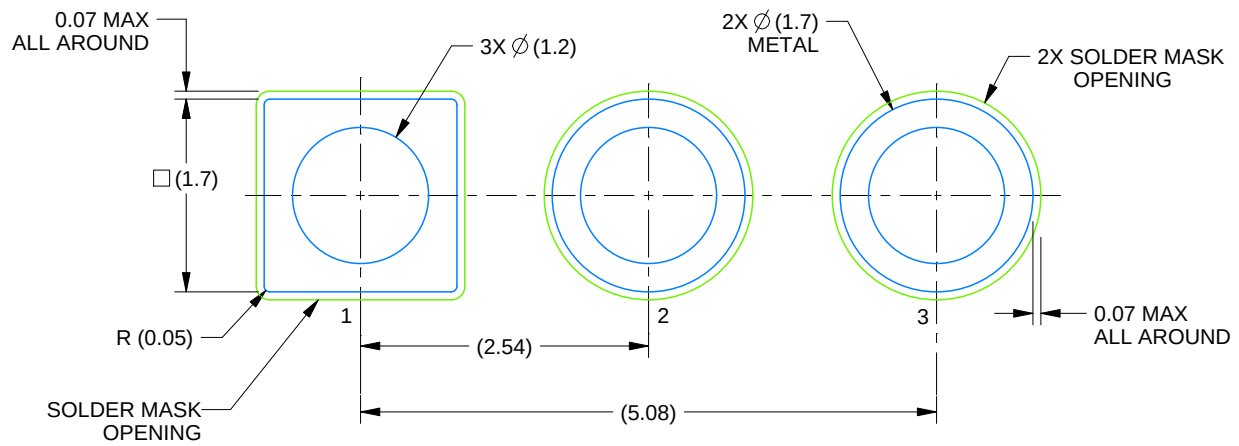
1. Dimensions are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC registration TO-220.

EXAMPLE BOARD LAYOUT

KCS0003B

TO-220 - 19.65 mm max height

TO-220



LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE:15X

4222214/B 08/2018

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